GREENSPARK HACKATHON

ERVEI

Predicting energy hardship before the heat arrives

Rowan, Sam, Chris, Shark

Payment difficulty is rising, and support is uneven

5 yrs

Residential customers with 90+ days of energy debt are at the highest level in five years

~1/3

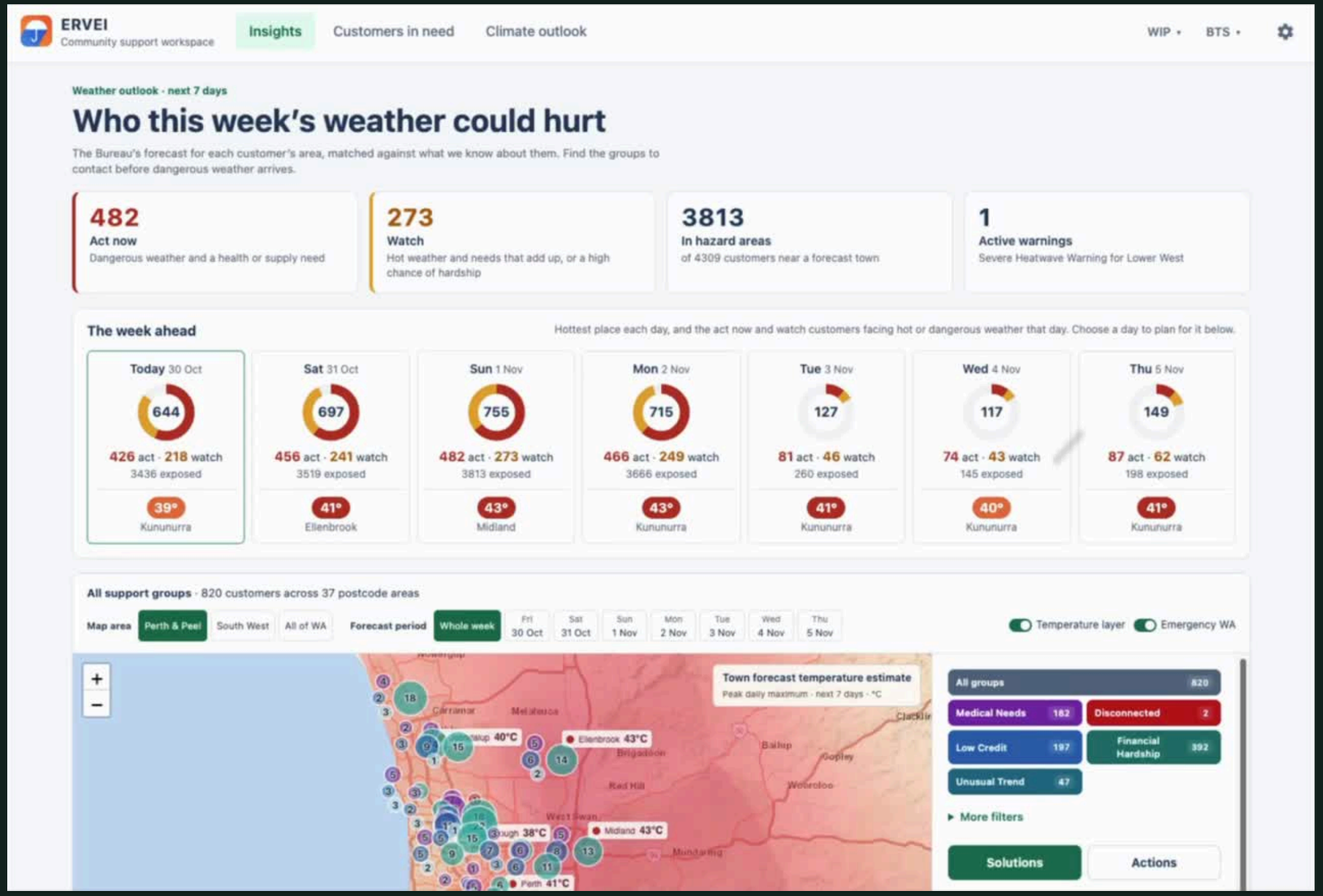
Of customers in debt did not get adequate support through a payment plan or hardship program

Early

New national rules push retailers to identify struggling customers and offer help before disconnection

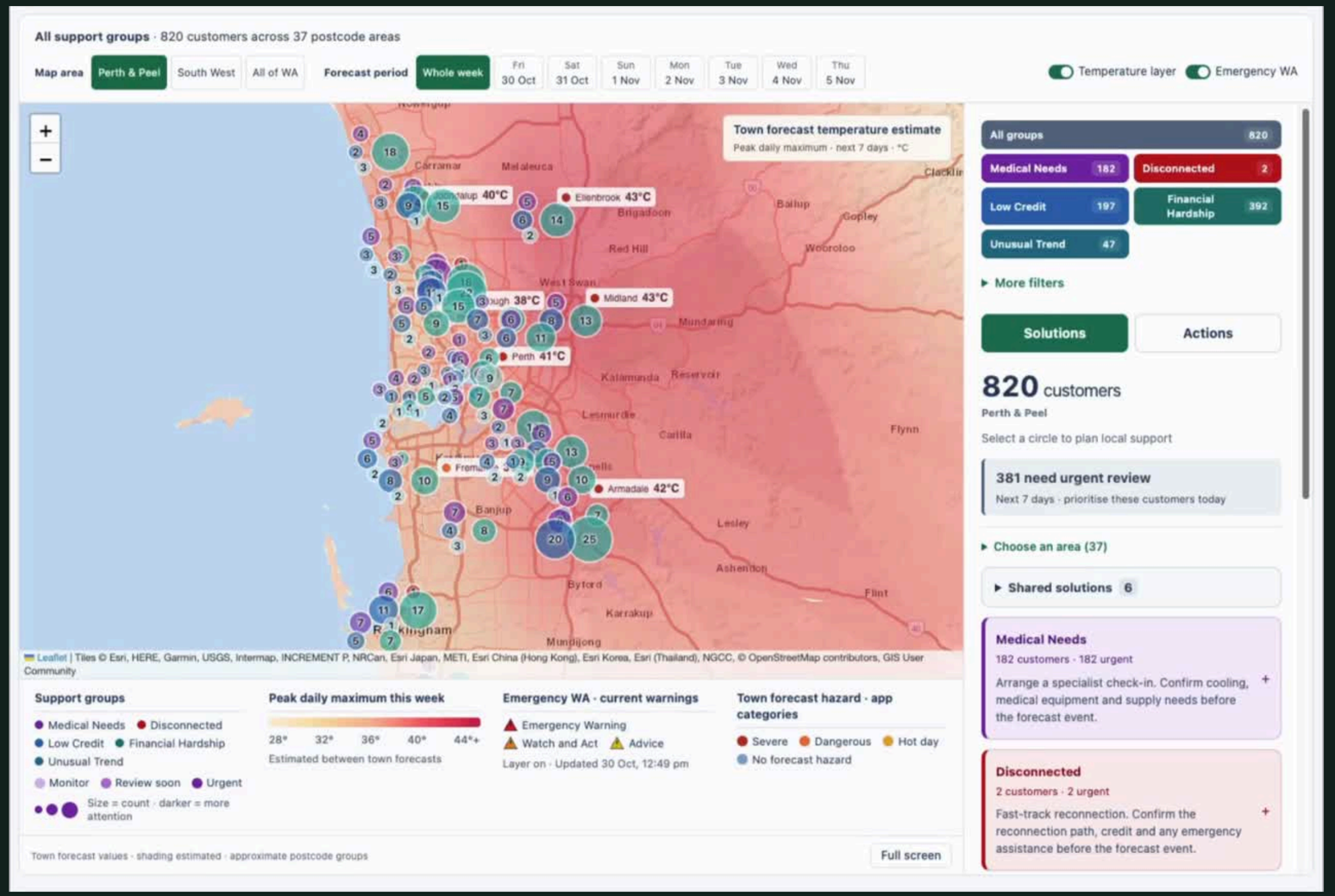
Source: AEMC media release, 10 September 2026 (AER data and review)

Demo



Simulated customer data. Real weather warnings and climate layers.

Demo


All support groups · 820 customers across 37 postcode areas
Map area
Perth & Peel
South West
All of WA
Forecast period
Whole week
Fri 30 Oct
Sat 31 Oct
Sun 1 Nov
Mon 2 Nov
Tue 3 Nov
Wed 4 Nov
Thu 5 Nov
Temperature layer
Emergency WA
Town forecast temperature estimate
Peak daily maximum · next 7 days · °C
Ellenbrook 43°C
Midland 43°C
Perth 41°C
Armadale 42°C
Leaflet | Tiles © Esri, HERE, Garmin, USGS, Intermap, INCREMENT P, NRCan, Esri Japan, METI, Esri China (Hong Kong), Esri Korea, Esri (Thailand), NGCC, © OpenStreetMap contributors, GIS User Community
Support groups
Medical Needs
Disconnected
Low Credit
Financial Hardship
Unusual Trend
Monitor
Review soon
Urgent
Size = count · darker = more attention
Peak daily maximum this week
28°
32°
36°
40°
44°+
Estimated between town forecasts
Emergency WA · current warnings
Emergency Warning
Watch and Act
Advice
Layer on · Updated 30 Oct, 12:49 pm
Town forecast hazard · app categories
Severe
Dangerous
Hot day
No forecast hazard
Town forecast values · shading estimated · approximate postcode groups
Full screen
All groups 820
Medical Needs 182
Disconnected 2
Low Credit 197
Financial Hardship 392
Unusual Trend 47
More filters
Solutions
Actions
820 customers
Perth & Peel
Select a circle to plan local support
381 need urgent review
Next 7 days · prioritise these customers today
Choose an area (37)
Shared solutions 6
Medical Needs
182 customers · 182 urgent
Arrange a specialist check-in. Confirm cooling, medical equipment and supply needs before the forecast event.
Disconnected
2 customers · 2 urgent
Fast-track reconnection. Confirm the reconnection path, credit and any emergency assistance before the forecast event.

Data sources

Customer Account

Personal details

Bills

Electricity consumption

Concession

Customer Survey and Call Transcripts

Additional details about their circumstances

Bureau of Meteorology

Weather forecast

Weather warnings

Fire danger ratings

Emergency WA

Emergency warnings

Climate Data

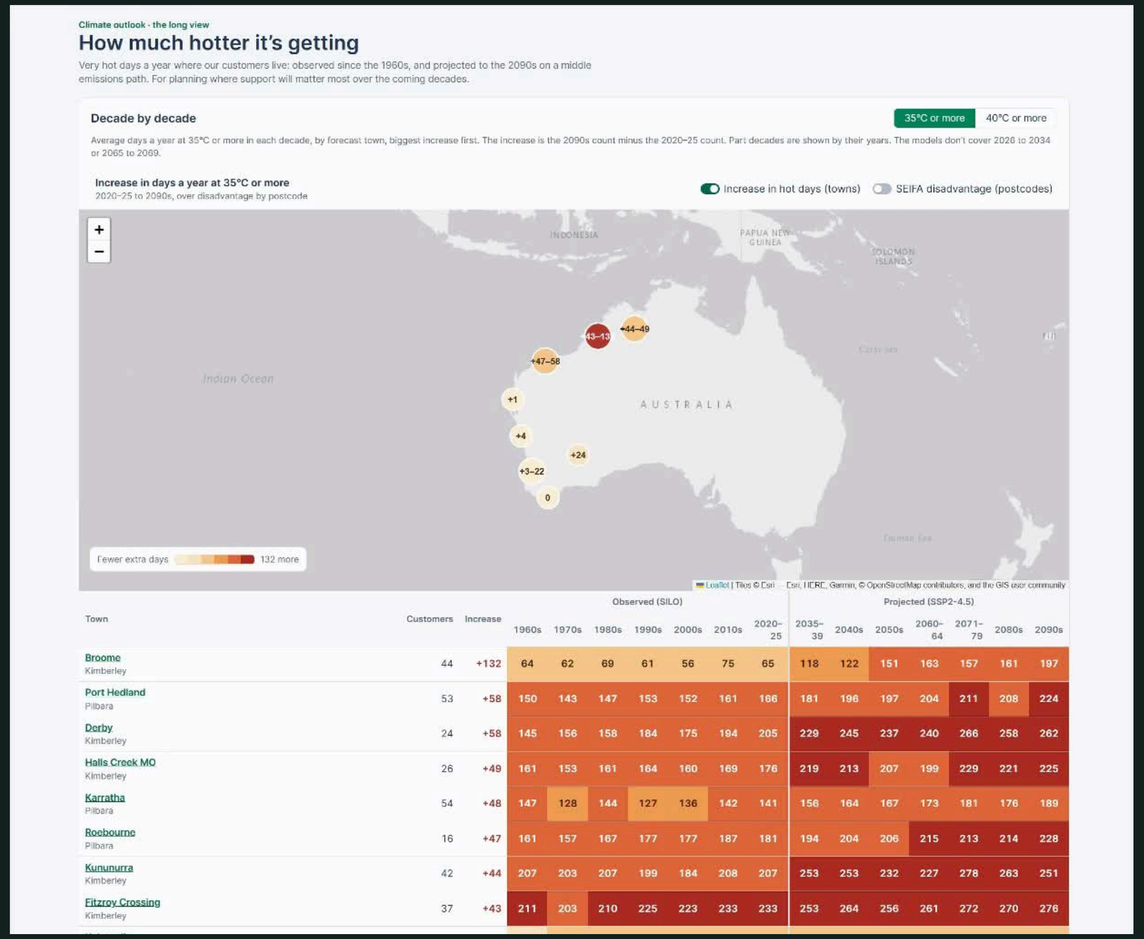
Past weather

Climate projections

ABS Data

Socioeconomic index (SEIFA)

Climate outlook



Climate outlook · the long view

How much hotter it's getting

Very hot days a year where our customers live: observed since the 1960s, and projected to the 2090s on a middle emissions path. For planning where support will matter most over the coming decades.

Decade by decade

35°C or more | 40°C or more

Average days a year at 35°C or more in each decade, by forecast town, biggest increase first. The increase is the 2090s count minus the 2020–25 count. Part decades are shown by their years. The models don't cover 2026 to 2034 or 2065 to 2069.

Increase in days a year at 35°C or more
2020–25 to 2090s, over disadvantage by postcode

Increase in hot days (towns) | SEIFA disadvantage (postcodes)

Town	Customers	Increase	Observed (SILO) 1960s	1970s	1980s	1990s	2000s	2010s	2020–25	Projected (SSP2-4.5) 2035–39	2040s	2050s	2060–64	2071–79	2080s	2090s
Broome Kimberley	44	+132	64	62	69	61	56	75	65	118	122	151	163	157	161	197
Port Hedland Pilbara	53	+58	150	143	147	153	152	161	166	181	196	197	204	211	208	224
Derby Kimberley	24	+58	145	156	158	184	175	194	205	229	245	237	240	266	258	262
Halls Creek MO Kimberley	26	+49	161	153	161	164	160	169	176	219	213	207	199	229	221	225
Karratha Pilbara	54	+48	147	128	144	127	136	142	141	156	164	167	173	181	176	189
Roebourne Pilbara	16	+47	161	157	167	177	177	187	181	194	204	206	215	213	214	228
Kununurra Kimberley	42	+44	207	203	207	199	184	208	207	253	253	232	227	278	263	251
Fitzroy Crossing Kimberley	37	+43	211	203	210	225	223	233	233	253	264	256	261	272	270	276

Climate projections and socioeconomic data

What we heard from Horizon Power

37,000

Residential customers

1,500

Prepaid accounts, mostly remote

3 yrs

Of payment history in their current algorithm

- Weather is not part of their existing hardship prediction
- Extreme weather response runs through disconnection moratoria and emergency credits
- Mentor feedback (Rennie): a tool like this is needed
- Different customer profile to Synergy (1.5M customers, large % in the metro area)

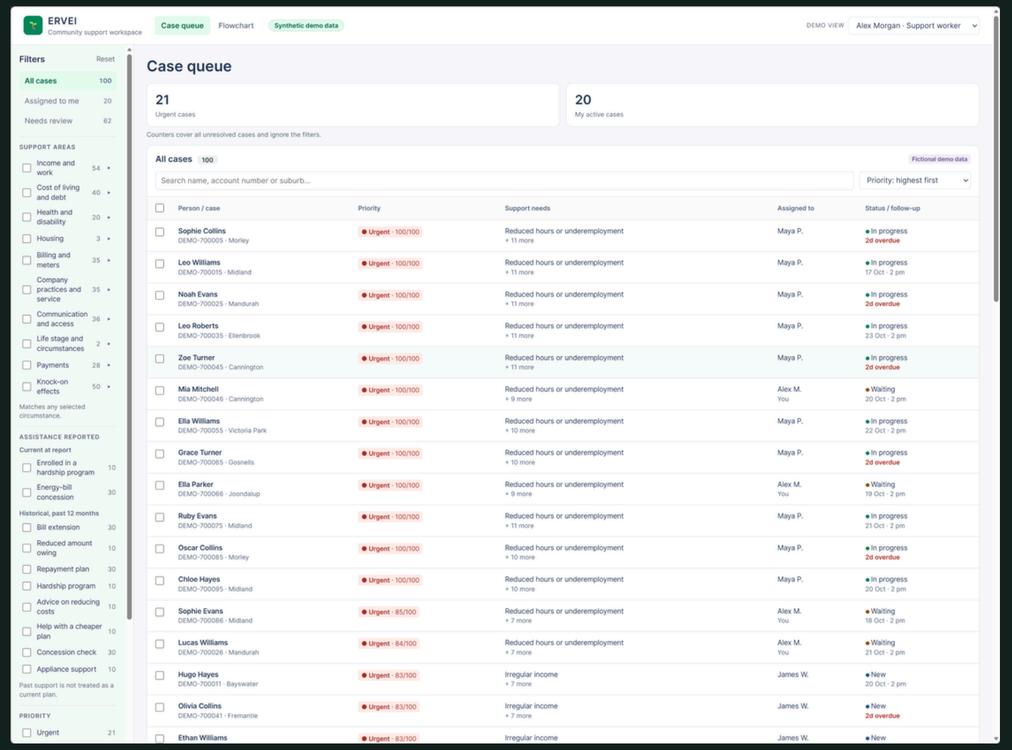


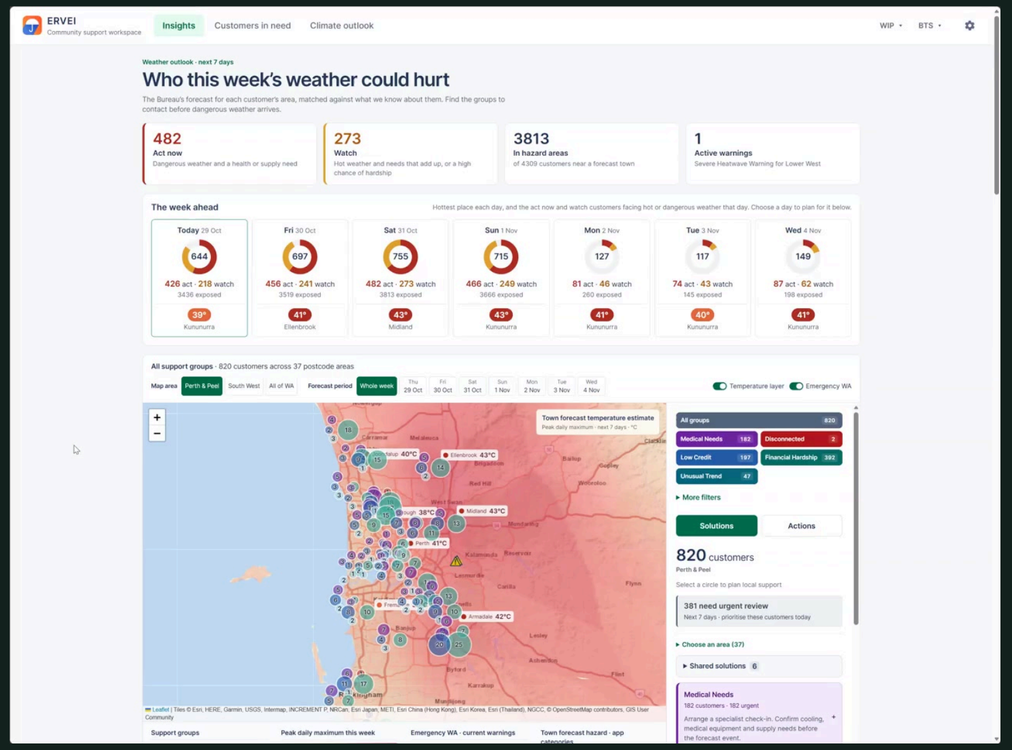




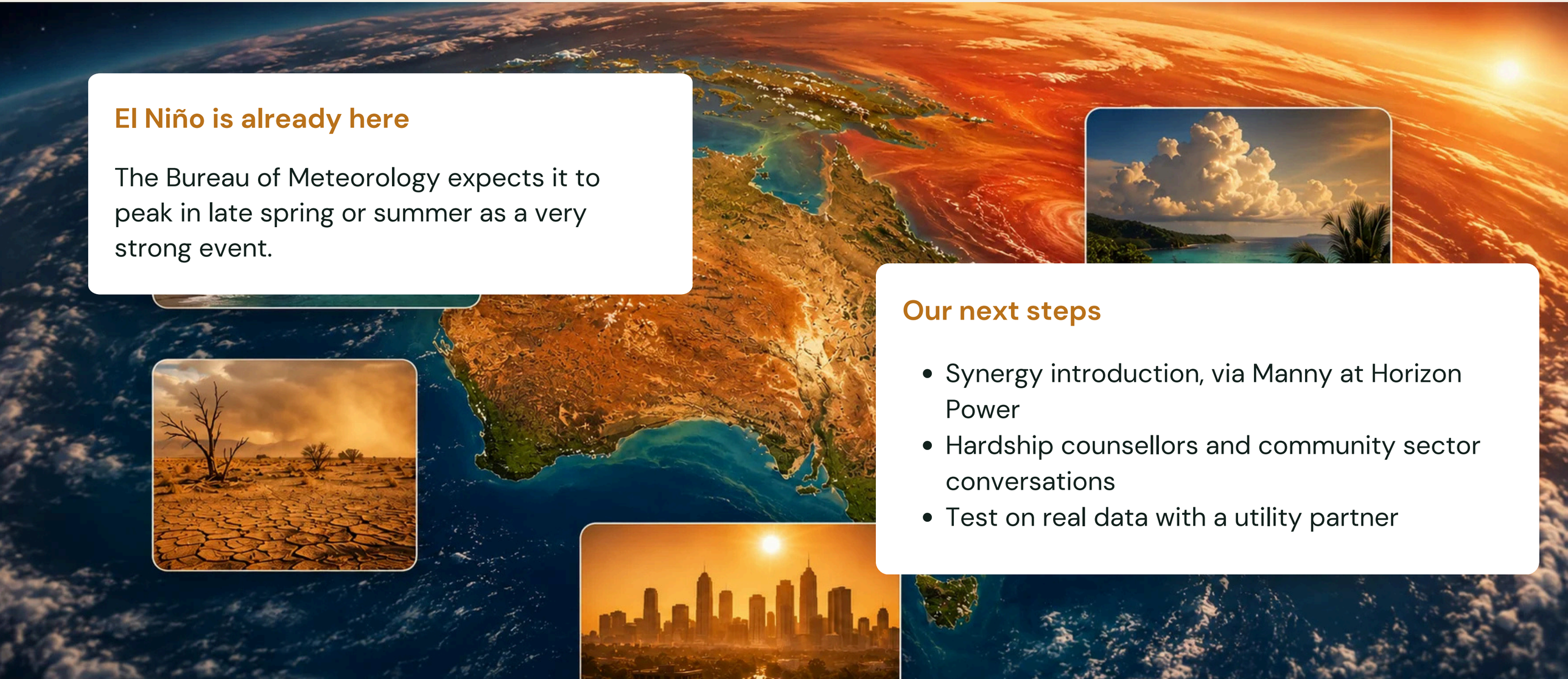


Pivot and changes





Why now, and what comes next



El Niño is already here

The Bureau of Meteorology expects it to peak in late spring or summer as a very strong event.

Our next steps

- Synergy introduction, via Manny at Horizon Power
- Hardship counsellors and community sector conversations
- Test on real data with a utility partner

Support before the bill, not after the disconnection

Live demo: ervei.fly.dev

Essential Services
Risk Of
Vulnerability:
Early Warning And
Intervention Tool